

Week 2:

- Start from *Iris* data, compare the *Sepal.Length* between different *Species*, to introduce how to test the **similarity** between chosen *Species* and baseline *Species*. And we have select 5 metrics (CDF/log-ratio/Jeffreys/DELR/Wald) in further study.

- Construct two simple model to predict *Iris*\$*Sepal.Length*:

M1:*Sepal.length*~*Sepal.width*

M2:*Sepal.length*~*Sepal.width*+*Species*(correct one)

Basis function		($y \ y^2$)	($y \log(y)$)	($y \ y^2 \log(y)$)↓
Accuracy based on different metrics	CDF	68.2%	68.4%	66.8%→
	Log-ratio	67.5%	67.8%	63.7%
	Jeffreys	68.4%	68.7%	60.7%↓
	DELR	68.6%	68.7%	58.6%↓
	Wald	68.3%	68.9%	67.0%→

All metrics are showing much the same accuracy while selecting.

In contrast, different metrics are showing different **sensitivity** while changing the **basis function**. However, ($y \log(y)$) is showing the best accuracy, while ($y \ y^2$) is also acceptable. But if we are not choosing the appropriate basis function, then the accuracy of **Jeffreys&DELR** would drop significantly, While **Wald&CDF** seems to be more stable.

*all accuracy are calculate through 1000 runs, where 95%CI $\pm 3.1\%$

*and mean accuracy is appoxiate value

- Simulation study on noraml data,where only use ($y \ y^2$):

1. Non-nested model:

M1: $Y \sim 3X1$

mean accuracy: 88.6%

M2: $Y \sim 2X2 + 1X3$

mean accuracy: 80.2%

2. Nested model:

M1: $Y \sim 3X1$

mean accuracy: 53.5%

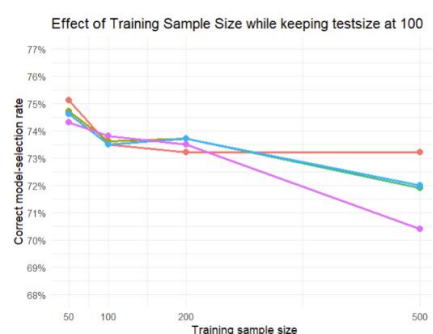
M2: $Y \sim 3X1 + 2X2 + 1X3$

mean accuracy: 85.4%

DRM is poor at selecting the correct model when the simpler one is correct, which is making sense as we are not penalizing complexity.

Sensitivity check

① (train/test size have no significant influence)



② Basis function

$$Y=1+3x_1+x_2+x_3$$

Basis function		(y)	(y y^2)	(y y^3)	(y y^2 y^3)
Accuracy based on different metrics	CDF	67.1%	73.9%	72.9%	78.1%
	Log-ratio	67.2%	74.6%	72.9%	77.7%
	Jeffreys	67.2%	74.9%	74.1%	78.2%
	DELR	67.2%	74.9%	74.0%	78.2%
	Wald	67.2%	75.0%	74.1%	78.3%

$$Y=1+3x_1$$

Basis function		(y)	(y y^2)	(y y^3)	(y y^2 y^3)
Accuracy based on different metrics	CDF	49.8%	48.3%	50.3%	49.3%
	Log-ratio	49.8%	48.9%	51.1%	50.8%
	Jeffreys	49.8%	49.6%	51.1%	49.9%
	DELR	49.8%	49.6%	51.1%	49.7%
	Wald	49.8%	49.6%	50.9%	49.9%

All metrics have much the **same accuracy** and show **similar sensitivity** towards basis function. However, for normal distributed data, the correct basis function should be (y y^2), but my result reveals that, which function have **best selection accuracy is depended on case**.

③ Effect size

The accuracy while selected a more-complex model will reduce if the **effect sizes of additional variables** become smaller.

Next step

I think there's no need to use all 5 metrics, if we want to compute more effectively. However I should check whether they do have same performance in more cases first. As the sensitivity check result in simulation study and Iris study (WALD/CDF more stable) is not the same, I will design a simulation study in 2 case: small sample&positively skewed distribution, to investigate where the non-consistent result is caused, so we can use the the less sensitive metric.